# Crime Count Forecasting per District

/ IT 462

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## DATASET INFORMATION

**Dataset Name:** Crimes - 2001 to Present
**Domain:** Public Safety / Urban Analytics
**Source:** City of Chicago Data Portal
**URL:** https://data.cityofchicago.org/Public-Safety/Crimes-2001-to-Present/ijzp-q8t2

**Approximate Dataset Size:**

- **Rows:** 971,865 records (2021-2024)
- **File Size:** 330 MB in CSV format
- **Number of Files/Tables:** 1 primary file

## MAIN ATTRIBUTES

The dataset contains **22 attributes** across the following types:

| Attribute Type | Attribute Name | Description | Example/Range |
|---|---|---|---|
| **Numeric** | ID | Unique record identifier | 12419690 |
| **Numeric** | Beat | Police beat number | 1-2535 |
| **Numeric** | District | Police district number | 1-31 |
| **Numeric** | Ward | City ward number | 1-50 |
| **Numeric** | Community Area | Community area code | 1-77 |
| **Numeric** | X Coordinate | State plane X coordinate | 1145000-1205000 |
| **Numeric** | Y Coordinate | State plane Y coordinate | 1850000-1950000 |
| **Numeric** | Latitude | GPS latitude coordinate | 41.6-42.1 |
| **Numeric** | Longitude | GPS longitude coordinate | -87.9 to -87.5 |
| **Categorical** | Primary Type | Main crime category | THEFT, BATTERY, ASSAULT, BURGLARY, ROBBERY (~30 categories) |
| **Categorical** | Description | Detailed crime subcategory | SIMPLE, AGGRAVATED, FROM BUILDING, etc. |
| **Categorical** | Location Description | Type of location where crime occurred | STREET, RESIDENCE, SIDEWALK, APARTMENT (~140 types) |
| **Categorical** | FBI Code | FBI crime classification code | 01A, 04B, 06, etc. |
| **Categorical** | IUCR | Illinois Uniform Crime Reporting code | 0460, 0820, 1310, etc. |

| | | | |
|---|---|---|---|
| **Categorical** | Arrest | Whether arrest was made | True/False |
| **Categorical** | Domestic | Whether incident was domestic-related | True/False |
| **Temporal** | Date | Incident occurrence timestamp | 01/01/2021 12:00:00 AM |
| **Temporal** | Year | Year of incident | 2021, 2022, 2023, 2024 |
| **Temporal** | Updated On | Last update timestamp | 02/08/2021 03:50:01 PM |
| **Text** | Case Number | Chicago Police Department case reference | JA123456 |
| **Text** | Block | Partially anonymized address | 001XX N STATE ST |
| **Text** | Description | Detailed crime description | SIMPLE, $500 AND UNDER, etc. |
| **Text** | Location | Combined lat/long in point format | (41.88, -87.63) |

*Table 1: Dataset attributes*

## PROBLEM STATEMENT

This project develops a regression-based crime count forecasting system that predicts the number of crime incidents in each Chicago district for future time periods using 971,865 historical crime records from 2021 to 2024. Instead of classifying locations as high risk or low risk, the system forecasts crime trends quantitatively at the district level using temporal and spatial patterns.

Accurate crime forecasting is important because it enables authorities to anticipate crime trends rather than react to past incidents. The system provides actionable insights for the Chicago Police Department, city officials, and urban planners to plan patrol deployments, allocate security resources, and evaluate policy effectiveness based on predicted crime volumes.
By predicting crime counts per district, the project supports proactive public safety planning, resource optimization, and evidence-based decision-making.

## MOTIVATION

Cities record millions of crime incidents each year, yet crime data is mostly used to describe past events rather than to anticipate future trends. Police departments typically respond after crimes occur because manually analyzing patterns across districts and time periods is impractical. This reactive approach limits efficient resource allocation and reduces the ability to prepare for fluctuations in crime levels.

The Chicago Crimes dataset contains large-scale historical records with spatial features (district, community area, coordinates), temporal features (hour, day, month, year), and detailed crime characteristics. These attributes enable the identification of temporal and geographic trends in crime activity over time.

Police departments require quantitative forecasts to support operational planning. Knowing approximately how many incidents to expect in each district during specific time periods allows for precise resource deployment, for example, anticipating 50 incidents versus 200 incidents in a district would significantly change staffing levels, patrol distribution, and response readiness. Numerical forecasting provides the granularity necessary for effective planning and evidence-based decision-making.

Therefore, this project aims to develop a district-level crime count forecasting system that transforms historical crime records into forward-looking insights, supporting proactive public safety planning rather than purely descriptive reporting.

## WHY THIS DATASET IS SUITABLE FOR SPARK AND THIS PROJECT

This dataset is ideal for Spark because its 971,865 records (330 MB) exceed the capacity of traditional single-machine processing tools and benefit from distributed computing. It includes rich spatial, temporal, and categorical features that support crime count forecasting per district over time. The dataset also introduces meaningful preprocessing challenges such as handling missing values, outliers, and encoding categorical variables. It enables efficient use of RDD operations and complex SQL queries for temporal pattern analysis and aggregation. Additionally, it supports regression modeling using Spark MLlib to predict future crime counts, demonstrating both Spark's technical capabilities and its practical application in addressing real-world public safety challenges. The dataset is publicly available, well-documented, and anonymized.